Teacher Wu Kuan | See It at a Glance #30 — Midpoint Triangle & Area Ratios

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1. The trap question

Join the midpoints of the three sides of a triangle. What fraction of the whole is the middle triangle?

2. The method

• Each midpoint segment is parallel to a side and exactly half its length (Midsegment Theorem).
• The four small triangles all have the same three side lengths, so they are congruent.
• So the middle triangle is 1/4 of the whole. Doing it twice gives $1/4 \times 1/4 = 1/16$.
• Similar figures: if side ratio is k, area ratio is k^2.

3. Parent check method

Ask your child to draw the four pieces and explain why they are identical (equal sides), not just say “1/4”.

Practice

1. Triangle ABC has area 48. Join the midpoints of its three sides. What is the area of the middle triangle?
2. Continuing, join the midpoints of that middle triangle. What is the area of the innermost triangle, and what fraction of the original is it?
3. In triangle ABC, D and E are midpoints of AB and AC, and BC = 14. Find DE, and the fraction of triangle ABC taken up by quadrilateral DBCE.
4. Two similar triangles have corresponding sides in the ratio 2 : 3. The smaller has area 20. What is the area of the larger?
5. Triangle ABC has area 36. D is on AB with AD : DB = 1 : 2. A line through D parallel to BC meets AC at E. Find the areas of triangle ADE and quadrilateral DBCE.

Answers

1. 12 ($48 \times 1/4$)
2. 3; it is 1/16 of the original ($1/4 \times 1/4$)
3. DE = 7; DBCE = $1 - 1/4 = 3/4$
4. 45 (area ratio 4 : 9)
5. ADE = 4 (ratio 1:3 → area 1:9); DBCE = 32

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